

Executive Overview: A production-grade, cross-platform mobile recipe discovery and meal management application built with **Flutter 3.22+**, **Firebase Cloud Firestore**, and **Provider** state management. Engineered strictly for Firebase Spark Plan efficiency without Firebase Storage or Cloud Functions dependencies, featuring real-time reactive streams, external HTTPS image delivery, role-based admin controls, and dynamic local state engines.

1. System Architecture & Admin Cloud Backend

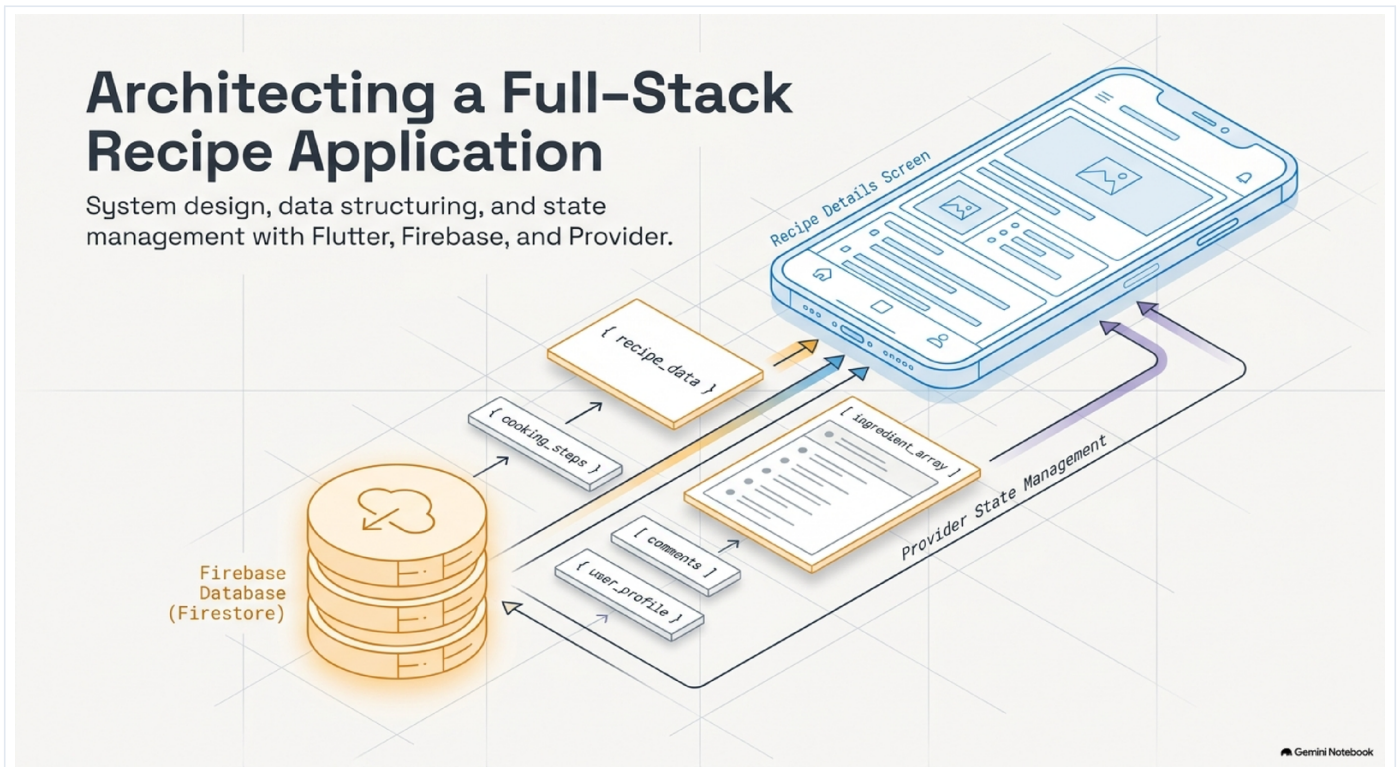


Figure 1: Full-Stack System Design & State Propagation Flow
Architectural blueprint showing Cloud Firestore collections, Provider state hydration, and reactive UI binding in Flutter.

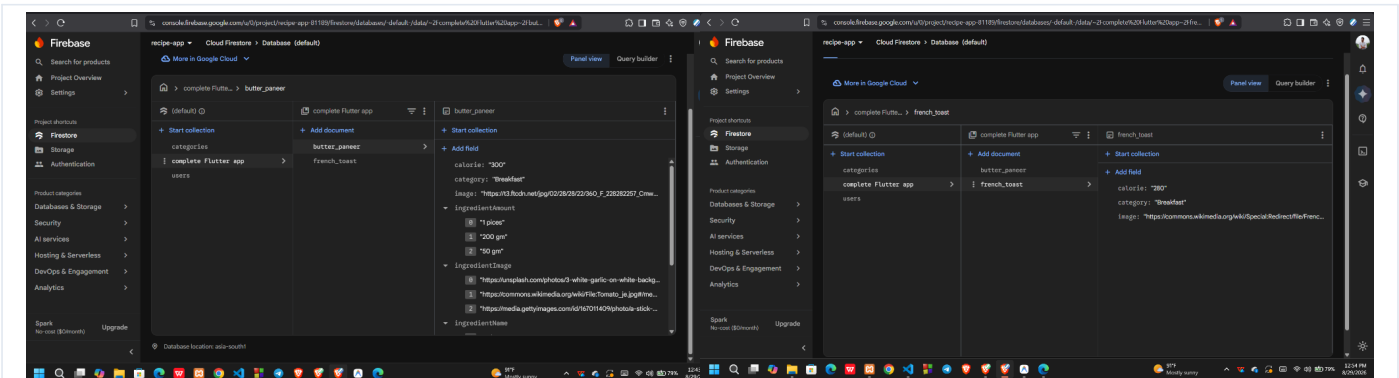
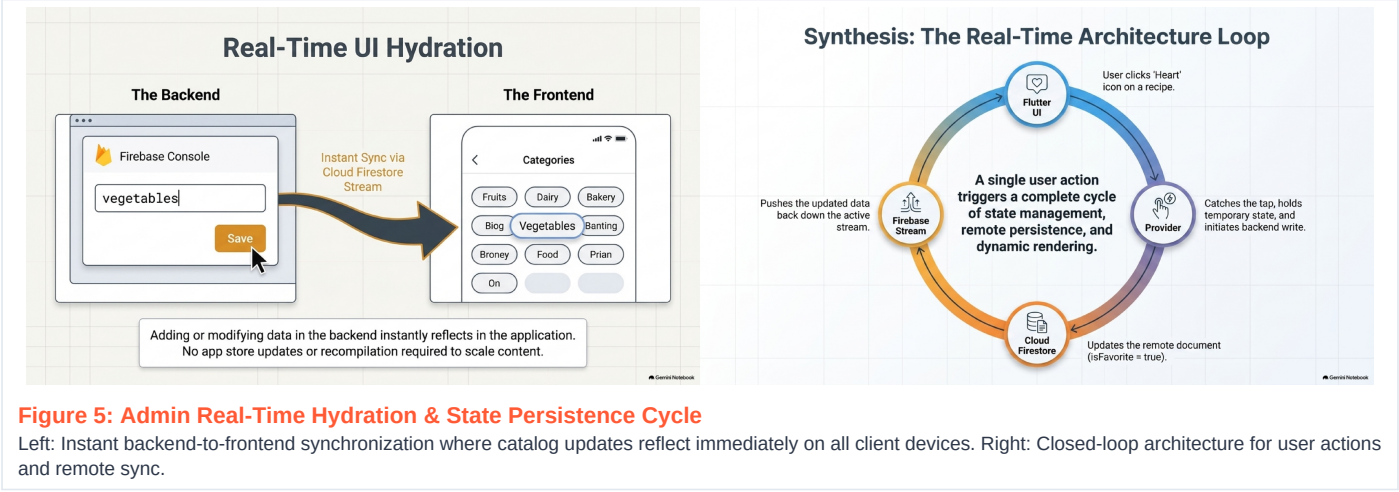
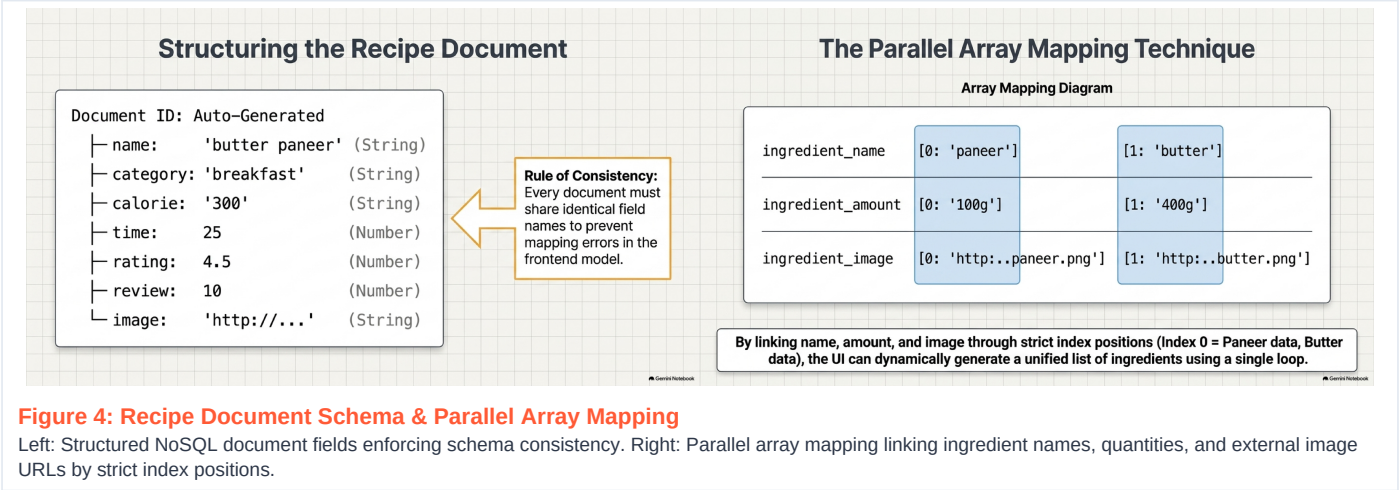
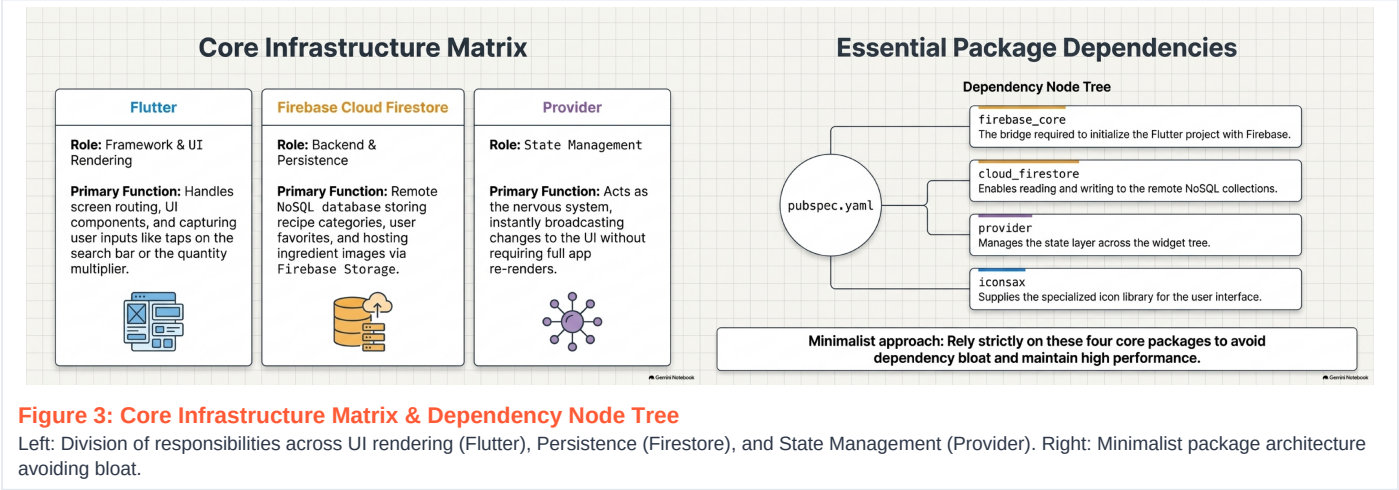


Figure 2: Cloud Firestore Database Hierarchy & Document Schemas
Live Firestore structure showing normalized collections (/categories, /recipes, /users) and structured recipe documents with validated HTTPS URLs.



2. Audience User Experience & Visual Interface

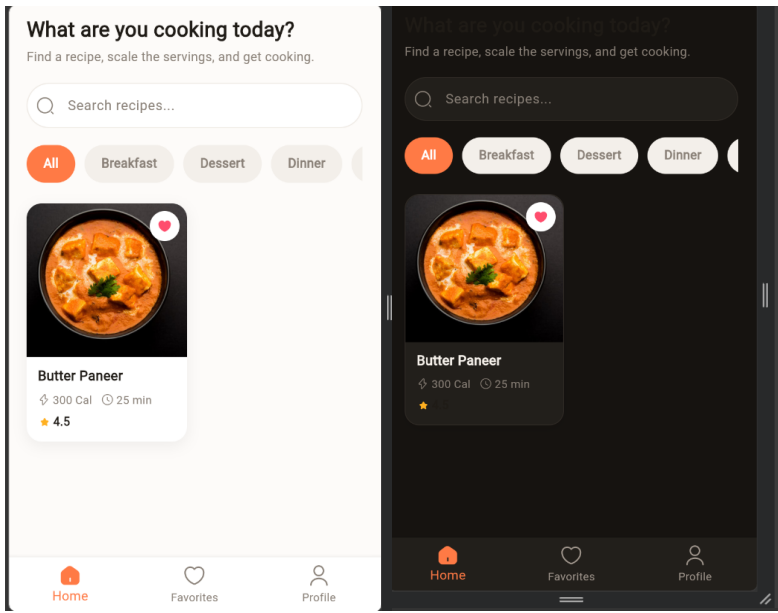


Figure 6: Interactive Home Discovery Screen (Adaptive Light & Dark Themes)

Audience discovery screen showcasing search bar, dynamic category chips, real-time recipe cards with calories, prep time, star ratings, and one-tap favorite toggling.

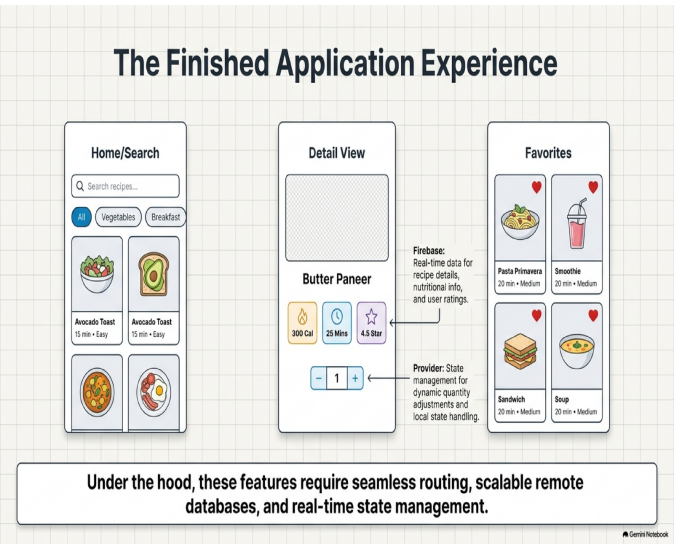
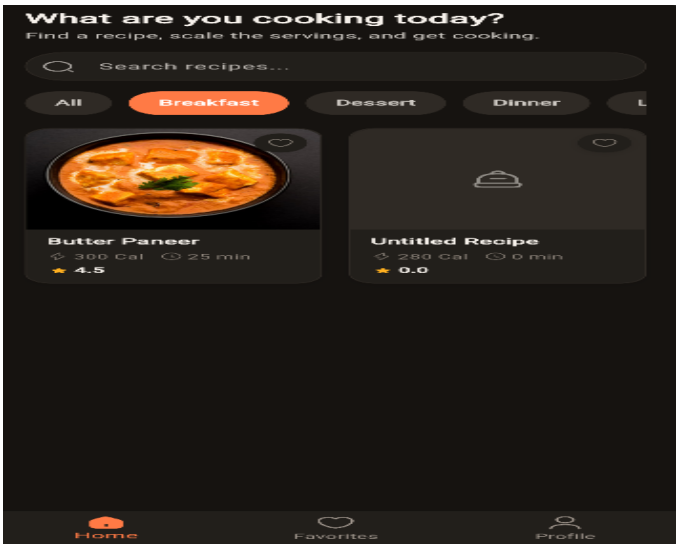


Figure 7: Dynamic Category Filtering & Finished Application Experience

Left: Instant category selection filtering live recipe streams. Right: Complete cross-screen interaction model spanning Home, Detail View, and User Favorites.

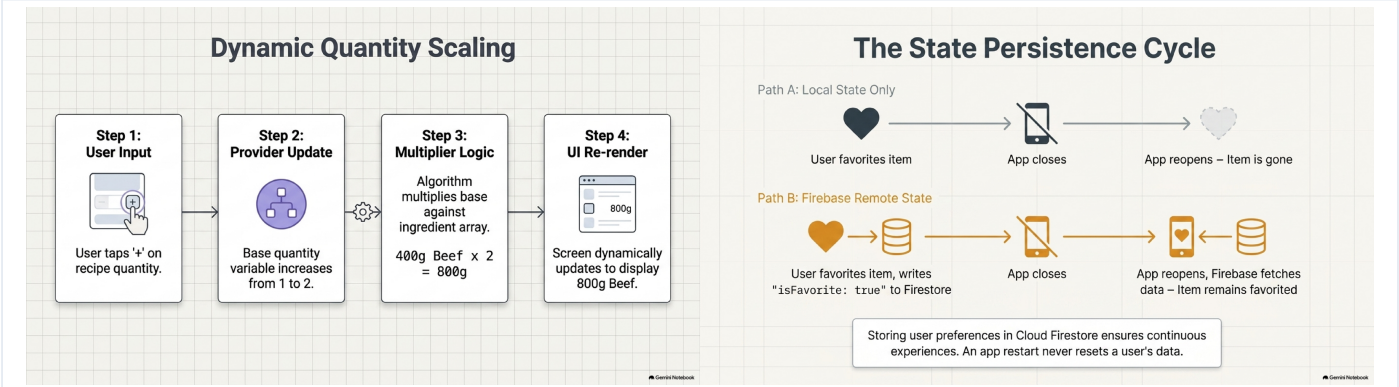


Figure 8: Dynamic Serving Multiplier Algorithm & State Persistence
Left: Mathematical scaling engine adjusting ingredient amounts in real-time when servings change. Right: Cloud-persisted state ensuring favorite meals survive app restarts.

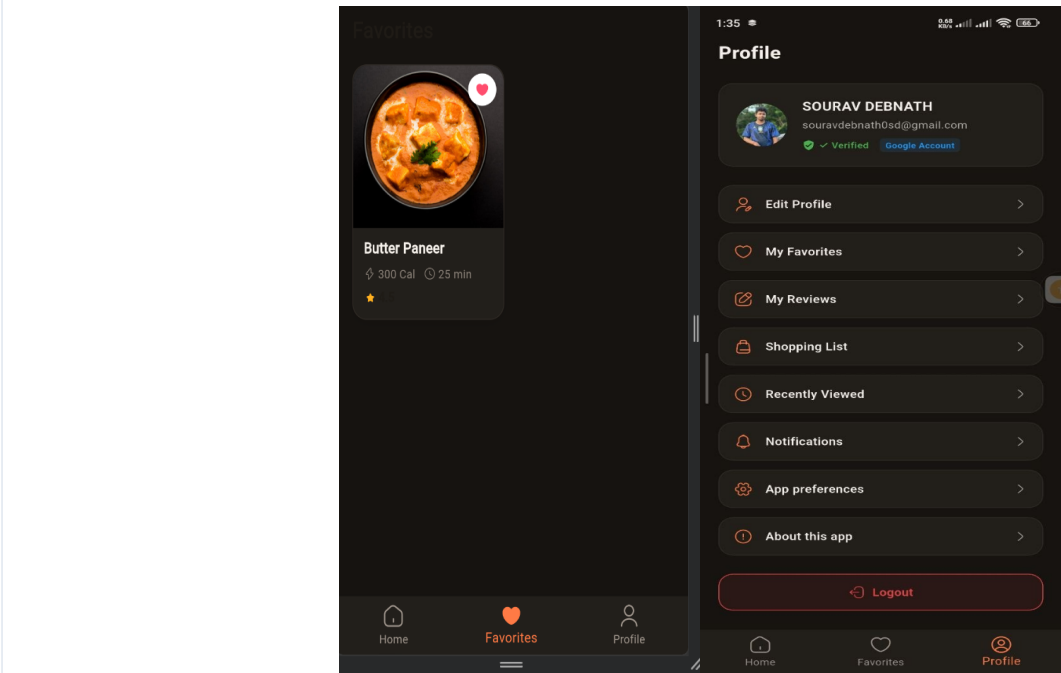


Figure 9: Synchronized Favorites & Verified User Profile Dashboard
Left: Real-time user favorites feed. Right: Complete profile management hub displaying verified Google Account status, navigation shortcuts for reviews, shopping lists, and preferences.

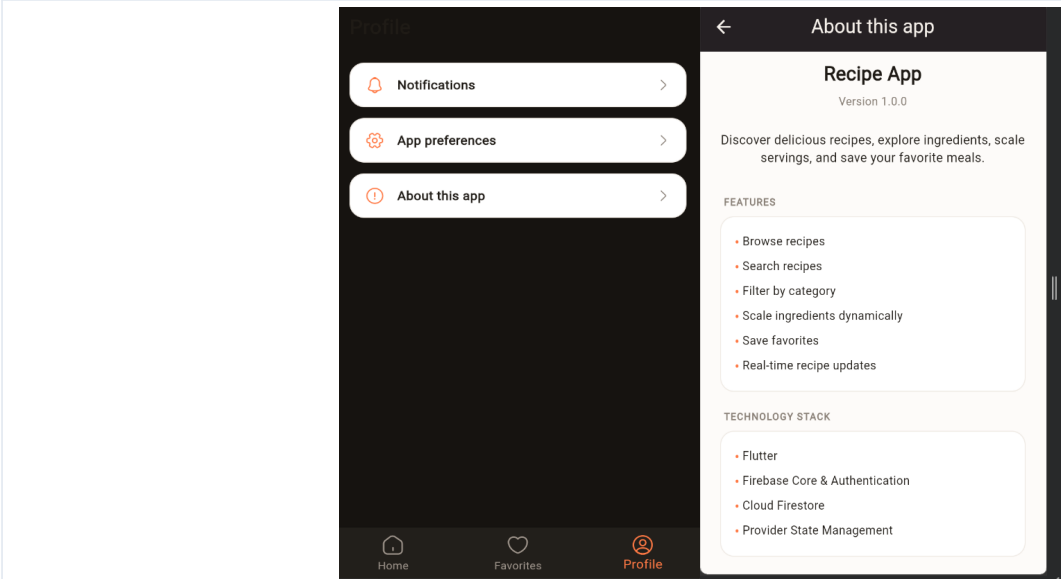


Figure 10: Dark Mode Profile Navigation & About Application Specifications
Left: Minimalist dark profile options. Right: In-app technical specifications, feature summary, and version metadata.

3. Technical Specifications & Quality Audit Matrix

Component / Area	Production Implementation Details	Status
Framework & SDK	Flutter 3.22+ with Dart 3 null-safety; responsive Android/Web/Windows layout	VERIFIED
State Management	Provider with decoupled RecipeProvider, AuthProvider, PreferencesProvider	VERIFIED
Cloud Database	Cloud Firestore real-time streams with offline cache enabled (Spark Plan compatible)	VERIFIED
Image Architecture	Validated external HTTPS URLs; zero Firebase Storage dependency; zero cost	VERIFIED
Security & Roles	Public recipe reading for audience; Admin custom claim required for write operations	VERIFIED
User Data Isolation	Subcollections (/users/{uid}/*) strictly isolated to authenticated user ID	VERIFIED
Automated Tests	54 unit and widget tests passing across models, providers, and navigation	54 / 54 PASS
Static Code Analysis	flutter analyze reports zero errors, zero warnings, and zero linter issues	0 ISSUES
Release Artifact	Android release APK compiled and available in release/app-release.apk (52.3 MB)	READY

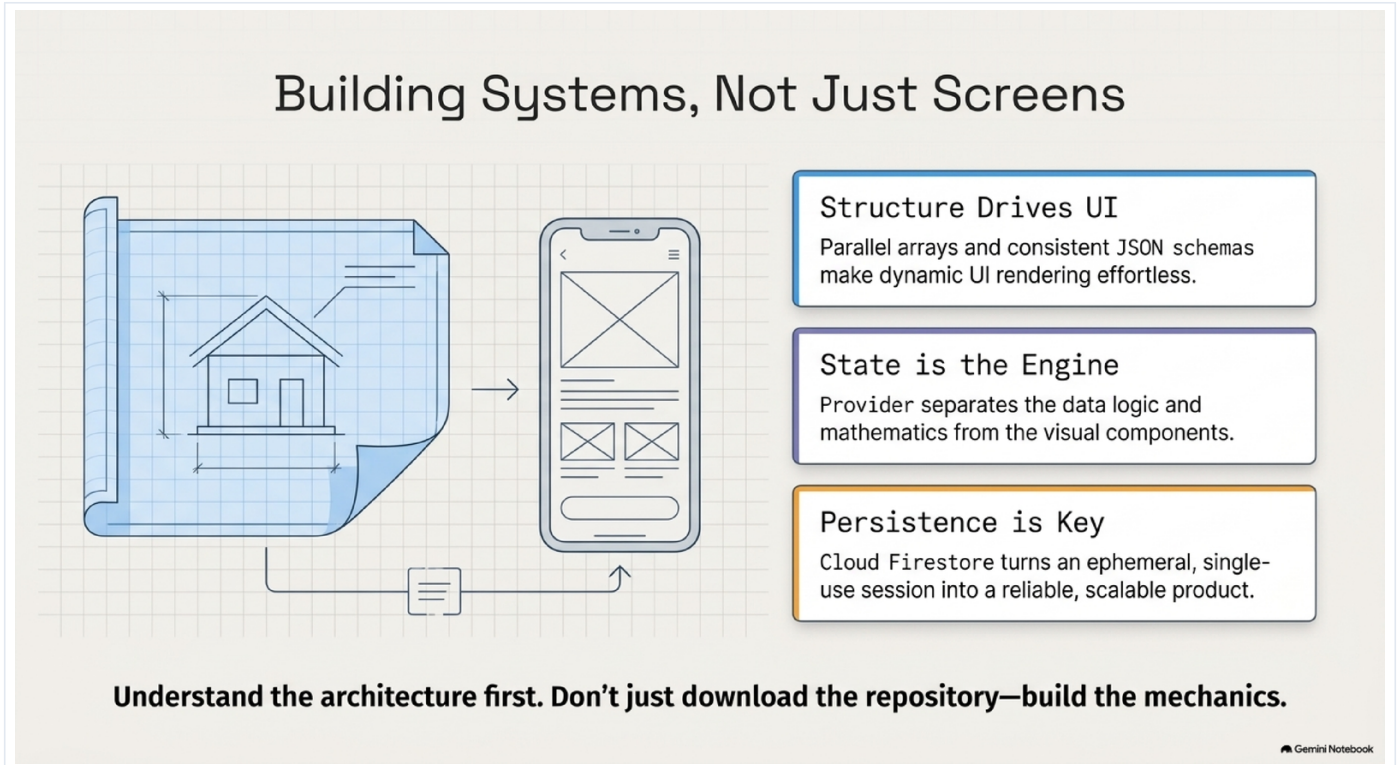


Figure 11: System Engineering Synthesis — Building Systems, Not Just Screens

Summary of system mechanics: schema-driven UI structure, Provider computational engine, and Cloud Firestore persistence delivering an enterprise-ready recipe platform.

- Repository & Distribution:**
- **GitHub Repository:** <https://github.com/SOURAVcse9/recipe-app-flutter-firebase>
 - **Production Release APK:** [release/app-release.apk](#)
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